Smooth-Copy of another model component: "scopy"

This model is a generalization of `copy`, please refer to `inla.doc("copy")` first.

This describes the way to copy another model component with an optional smooth/spline scaling, like with

$$\eta = u + v$$

where v is a smooth copy of u (component-wise)

$$v = \beta(z) \times \text{copy}(u)$$

where $\beta(z)$, a smooth/spline function of the covariate z. The smooth scaling is done **component-wise** for u, so if u are defined with domain $(1, 2, \ldots, m)$, i.e. $u = (u_1, u_2, \ldots, u_m)$, then z must be $z = (z_1, z_2, \ldots, z_m)$, so that

$$v_i = \beta(z_i)u_i, \qquad i = 1, 2, \ldots, m.$$

Hyperparameters

The hyperparameters are the value of the spline at n fixed (equally distant) locations, (l_i, β_i), for $i = 1, \ldots, n$. Let $l_M = (l_n + l_1)/2$ be the mid-points of the locations and $l_L = (l_n - l_1)$ its length. These β-parameters defines the interpolating spline (of second order). Let Q be the scaled-precision matrix for RW2 and define the eigen decomposition, as

$$Q = \sum_{i=1}^{n} \lambda_i v_i v_i^T$$

where $\lambda_{n-1} = \lambda_n = 0$, assuming descreasing eigenvalues. We can use $v_n = (1, \ldots, 1)^T$ and $v_{n-1} = (-0.5, \ldots, 0.5)^T$. Define the matrix with scaled v_i' as columns,

$$W = \left[v_n^T \;\middle|\; v_{n-1}^T \;\middle|\; \tfrac{1}{\sqrt{\lambda_{n-2}}} v_{n-2}^T \;\middle|\; \cdots \;\middle|\; \tfrac{1}{\sqrt{\lambda_1}} v_1^T \right].$$

Let the columns vectors of W be $w_1, w_2, \ldots$. We parameterize the spline at locations $l_1, \ldots, l_n$, as $\beta = (\beta_1, \ldots, \beta_n)$, as

$$\beta = \sum_{i=1}^{n} \theta_i w_i, \qquad i = 1, \ldots, n,$$

In this way, θ_1 is the overall mean, θ_2 is the (dimension-less) slope, and $\theta_3, \ldots, \theta_n$ are weights for the basis-function expansion of the spline that is beyond the constant and the linear term.

Since Q is scaled, then using independent $N(0, 1)$ prior for each $\theta_3, \ldots, \theta_n$ will make the prior deviation from the mean and slope, also $N(0, 1)$ (in an average sense). So only one common scaling of the prior precisions for these parameters of this prior are needed to shrink it more, or to shrink it less. For this reason the prior for θ_j, $j = 3, \ldots, n$, is *defined* to be the same as the prior for θ_3, hence only the prior for θ_3 needs to be specified.

Doing this from within `R`, we can evaluate the spline at any point within $[l_1, l_n]$, we can use

```
sfun <- splinefun(loc, beta, method = "natural")
new.value <- sfun(new.loc)
```

The functions `inla.scopy.summary` and `INLA:::inla.scopy.define` can be consulted for further details.

We can control n and the covariate with `control.scopy` within `f()`,

```
control.scopy = list(
covariate = ...,
n = 9)
```

where

covariate gives the covariate that is used

n is the number of hyperparameters used in the spline ($n = 2$ or $5 \leq n \leq 15$).

The `f()`-argument `precision`, defines how close the copy is, is similar as for model `copy`.

The priors for the mean, slope and the deviation from them, are given by the `hyper`-argument. See also the example.

Spesification

doc `Create a scopy of a model component`

hyper

- **theta1**
  - **hyperid** `36101`
  - **name** `mean`
  - **short.name** `mean`
  - **initial** `1`
  - **fixed** `FALSE`
  - **prior** `normal`
  - **param** `1 10`
  - **to.theta** `function(x) x`
  - **from.theta** `function(x) x`
- **theta2**
  - **hyperid** `36102`
  - **name** `slope`
  - **short.name** `slope`
  - **initial** `0`
  - **fixed** `FALSE`
  - **prior** `normal`
  - **param** `0 10`
  - **to.theta** `function(x) x`
  - **from.theta** `function(x) x`
- **theta3**
  - **hyperid** `36103`
  - **name** `spline.theta1`
  - **short.name** `spline`
  - **initial** `0`
  - **fixed** `FALSE`
  - **prior** `laplace`
  - **param** `0 10`
  - **to.theta** `function(x) x`
  - **from.theta** `function(x) x`
- **theta4**
  - **hyperid** `36104`
  - **name** `spline.theta2`
  - **short.name** `spline2`
  - **initial** `0`
  - **fixed** `FALSE`
  - **prior** `none`
  - **param**
  - **to.theta** `function(x) x`

from.theta `function(x) x`

theta5

hyperid `36105`
name `spline.theta3`
short.name `spline3`
initial `0`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta6

hyperid `36106`
name `spline.theta4`
short.name `spline4`
initial `0`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta7

hyperid `36107`
name `spline.theta5`
short.name `spline5`
initial `0`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta8

hyperid `36108`
name `spline.theta6`
short.name `spline6`
initial `0`
fixed `FALSE`
prior `none`
param
to.theta `function(x) x`
from.theta `function(x) x`

theta9

hyperid `36109`
name `spline.theta7`

short.name spline7
initial 0
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta10

hyperid 36110
name spline.theta8
short.name spline8
initial 0
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta11

hyperid 36111
name spline.theta9
short.name spline9
initial 0
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta12

hyperid 36112
name spline.theta10
short.name spline10
initial 0
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta13

hyperid 36113
name spline.theta11
short.name spline11
initial 0
fixed FALSE
prior none

param

to.theta `function(x) x`

from.theta `function(x) x`

theta14

hyperid `36114`

name `spline.theta12`

short.name `spline12`

initial `0`

fixed `FALSE`

prior `none`

param

to.theta `function(x) x`

from.theta `function(x) x`

theta15

hyperid `36115`

name `spline.theta13`

short.name `spline13`

initial `0`

fixed `FALSE`

prior `none`

param

to.theta `function(x) x`

from.theta `function(x) x`

constr `FALSE`

nrow.ncol `FALSE`

augmented `FALSE`

aug.factor `1`

aug.constr

n.div.by

n.required `FALSE`

set.default.values `FALSE`

pdf `scopy`

Example 1

```
if (FALSE) {
    inla.setOption(smtp = 'taucs', safe = FALSE, num.threads = "1:1")
}

N <- 200
s <- 0.1
x <- 1:N
eta <- 1 + x / N + sin(x * 0.1) * exp(-2*x/N)
y <- eta + rnorm(N, sd = s)
m <- 15

Y <- matrix(NA, N + 1, 2)
Y[1:N, 1] <- y
Y[N+1, 2] <- mean(y)
r <- inla(Y ~ -1 +
              ## this model will just define the 'overall level',
              ## but with one value for each i.
              ## We need this as as can then scale this one with
              ## the spline. We add a point with
              ## the second likelihood to lock-it in place
              f(idx,
                model = "rw1",
                scale.model = TRUE,
                constr = FALSE,
                values = 1:N,
                hyper = list(prec = list(initial = 15,
                                         fixed = TRUE))) +
              ## the 'overall level' is scaled by a spline
              f(idx.scopy, scopy = "idx",
                hyper = list(mean = list(param = c(1, 0.1)),
                             slope = list(param = c(0, 0.1)),
                             spline = list(param = c(0, 20))),
                control.scopy = list(covariate = x, n = m)),
          ##
          data = list(idx = c(rep(NA, N), 1),
                      idx.scopy = c(1:N, NA),
                      x = c(x,1),
                      m = m),
          family = c("gaussian", "gaussian"),
          control.family = list(
              list(hyper =
                       list(prec = list(
                                initial = log(1/s^2),
                                fixed = TRUE))),
              list(hyper =
                       list(prec = list(
                                initial = 15,
                                fixed = TRUE)))),
          control.compute = list(config = TRUE, residuals = TRUE))

plot(x, y, pch = 19)
## note that the locations are not stored in the results, hence
## we can set them here. This is just for the ease of the output,
## the results are unchanged.
beta <- inla.scopy.summary(r, "idx.scopy", range = c(1, N))
```

```
s.mean <- mean(r$summary.random$idx$mean)
lines(beta$x, s.mean * (beta$mean), lwd = 3, col = "blue")
lines(beta$x, s.mean * (beta$mean + 2 * beta$sd), lwd = 2,
      lty = 2, col = "black")
lines(beta$x, s.mean * (beta$mean - 2 * beta$sd), lwd = 2,
      lty = 2, col = "black")
```

Example 2

```
if (FALSE) {
    inla.setOption(smtp = 'taucs', safe = FALSE, num.threads = "1:1")
}
N <- 100
s <- 0.5
x <- 1:N
eta <- 1 + x / N
y <- eta + rnorm(N, sd = s)
m <- 2

Y <- matrix(NA, N + 1, 2)
Y[1:N, 1] <- y
Y[N+1, 2] <- mean(y)
r <- inla(Y ~ -1 +
              ## this model will just define the 'overall level',
              ## but with one value for each i.
              ## We need this as as can then scale this one with
              ## the spline. We add a point with
              ## the second likelihood to lock-it in place
              f(idx,
                model = "rw1",
                scale.model = TRUE,
                constr = FALSE,
                values = 1:N,
                hyper = list(prec = list(initial = 15, fixed = TRUE))) +
              ## the 'overall level' is scaled by a spline
              f(idx.scopy, scopy = "idx",
                hyper = list(mean = list(param = c(1, 0.1)),
                             slope = list(param = c(0, 0.1)),
                             spline = list(param = c(0, 20))),
                control.scopy = list(covariate = x, n = m)),
          ##
          data = list(idx = c(rep(NA, N), 1),
                      idx.scopy = c(1:N, NA),
                      x = c(x,1),
                      m = m),
          family = c("gaussian", "gaussian"),
          control.family = list(
              list(hyper = list(prec = list(
                                    initial = log(1/s^2),
                                    fixed = TRUE))),
              list(hyper = list(prec = list(
                                    initial = 15,
                                    fixed = TRUE)))),
          control.compute = list(config = TRUE, residuals = TRUE))

plot(1:N, y, pch = 19, main = "SCOPY")
```

```
## note that the locations are not stored in the results, hence
## we can set them here. This is
## just for the ease of the output, the results are unchanged.
beta <- inla.scopy.summary(r, "idx.scopy", range = c(1, N))
s.mean <- mean(r$summary.random$idx$mean)
lines(beta$x, s.mean * (beta$mean), lwd = 3, col = "blue")
lines(beta$x, s.mean * (beta$mean + 2 * beta$sd), lwd = 2, lty = 2, col = "black")
lines(beta$x, s.mean * (beta$mean - 2 * beta$sd), lwd = 2, lty = 2, col = "black")

## this is just a check
rr <- inla(y ~ 1 + x,
           data = data.frame(y, x),
           control.family = list(hyper = list(prec = list(
                                                 initial = log(1/s^2),
                                                 fixed = TRUE))))
inla.dev.new()
plot(1:N, y, pch = 19, main = "y ~ 1 + x")
lines(1:N, rr$summary.linear.predictor$mean, lwd = 3, col = "blue")
lines(1:N, rr$summary.linear.predictor$"0.025quant", lwd = 2,
      lty = 2, col = "black")
lines(1:N, rr$summary.linear.predictor$"0.975quant", lwd = 2,
      lty = 2, col = "black")
```

Notes

This model is experimental.